(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/09/2026
(21) Application No.202641111370 A
(43) Publication Date : 25/09/2026

(54) Title of the invention : AI-Driven Context-Aware System and Method for Centralized Linux Desktop Environment Management, Adaptive Security Enforcement, and Dynamic Workspace Transformation

(51) International classification	:H04L 29/06, H04L 9/40, G06F 21/57, H04L 12/24, G06F 21/55
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dayananda Sagar College of Engineering
Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India
(72)**Name of Inventor :**
1)Surendra S
2)Dr. Pallavi Kulkarni
3)Dr. Mohammad Tajuddin

(57) Abstract :
The present invention discloses an AI-driven context aware system and method for centralized Linux desktop environment management, adaptive security enforcement and dynamic workspace transformation designed to enhance security, operational efficiency, and administrative control in enterprise, cloud, and critical infrastructure environments. Conventional Linux desktop management systems primarily rely on static user profiles, predefined access permissions, and login-time policy enforcement, making them inadequate for addressing evolving cyber threats, insider attacks, dynamic organizational requirements, and Zero Trust security models. The proposed invention introduces a centralized intelligent management framework comprising a profile repository, a context-aware policy decision engine, an AI-based behavioural analytics module, an adaptive workspace transformation engine, a dynamic privilege management module, a continuous security enforcement engine, and a tamper-evident audit subsystem. The framework continuously evaluates user identity, device posture, network trust level, geographic location, time, organizational role, and real-time behavioural patterns to compute a dynamic trust score. Based on this assessment, the system autonomously transforms the Linux desktop environment by modifying applications, desktop layouts, access privileges, peripheral device availability, network permissions, and security configurations without requiring user logout or administrator intervention. The invention further provides continuous Zero Trust verification, autonomous threat response, self-healing desktop recovery, and cryptographically protected audit logging to ensure integrity, compliance, and resilience against cyberattacks. The proposed architecture supports multiple Linux distributions and desktop environments through a unified policy framework, enabling adaptive workspace personalization while maintaining stringent security controls. By integrating artificial intelligence, contextual awareness, dynamic policy orchestration, and autonomous security enforcement into a single platform, the invention significantly improves enterprise Linux desktop management, reduces administrative overhead, strengthens cybersecurity, and provides a scalable solution for secure digital workspaces in government, defense, healthcare, education, financial institutions, and other critical infrastructure sectors.

No. of Pages : 26 No. of Claims : 10